Reverse-Complement Aware Dilated CNN–BiGRU Architecture for Regulatory Genomic Sequence Modeling

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Abstract

Predicting regulatory activity from nucleotide sequence is a central problem in computational genomics, with applications in chromatin accessibility, transcription factor binding, and gene regulation. In this work, we present a modular deep learning framework for regulatory sequence prediction and evaluate a hybrid architecture designed to combine complementary genomic modeling principles. The proposed model integrates reverse-complement-aware processing, dilated residual convolutional blocks, bidirectional recurrent modeling, and gated attention pooling.

We evaluate the approach on two tasks: ATAC-seq chromatin accessibility prediction and CTCF binding prediction. All models are trained and evaluated under the same pipeline, using chromosome-based splits, one-hot encoded DNA sequences, improved negative sampling, and consistent optimization settings. We compare against classical baselines, including logistic regression, multilayer perceptrons, and CNNs, as well as SOTA-inspired architectures based on DeepSEA, DanQ, and Basenji.

Across both tasks, the proposed model achieves the strongest validation performance. The improvement is particularly large on the more challenging ATAC task, while CTCF results show that the model also remains competitive in a motif-driven regulatory setting. Overall, the results suggest that combining strand-aware processing, long-range convolutional context, recurrent sequence modeling, and attention-based aggregation provides an effective inductive bias for nucleotide sequence-based regulatory prediction

Keywords

genomic sequence modeling, regulatory DNA prediction, reverse-complement, invariance, deep learning, convolutional neural networks, recurrent neural networks, attention mechanisms, ATAC-seq, CTCF binding prediction

1 Introduction

DNA is a nucleotide acid sequence that encodes not only protein-coding information, but also regulatory signals controlling when and where genes are expressed. Identifying these regulatory signals from sequence is a major problem in computational genomics. Regulatory regions influence chromatin accessibility, transcription factor binding, cell-type-specific activity, and disease-associated genetic variation. As a result, models that can predict regulatory activity directly from DNA sequence are valuable for both biological interpretation and biomedical applications.

In this work, we focus on two representative regulatory prediction tasks. The first is ATAC-seq chromatin accessibility prediction [1], where the goal is to identify DNA regions likely to be accessible to regulatory machinery. The second is CTCF binding prediction [4], where the goal is to predict binding regions of a specific transcription factor. Both tasks can be formulated as binary sequence classification problems: given a fixed-length DNA window, the model predicts whether the region corresponds to a positive regulatory signal or background sequence.

Deep learning is well suited to this setting because DNA contains patterns at multiple scales. Short sequence motifs can indicate transcription factor binding, while broader sequence context may influence accessibility and regulatory activity. In addition, DNA is double-stranded, meaning that regulatory patterns may appear on either the forward strand or the reverse-complement strand [6]. A strong model for regulatory sequence prediction should therefore combine local motif detection, larger contextual modeling, and strand-aware processing.

Several existing genomic deep learning models exploit some of these ideas, but they often emphasize different aspects of the problem: convolutional models focus on motif discovery, recurrent models capture positional dependencies, and dilated architectures increase receptive field size. Our goal is to study whether combining these complementary inductive biases in a single controlled framework improves regulatory prediction across different biological tasks.

We propose a modular deep learning framework for nucleotide sequence-based regulatory prediction. The framework includes reusable components for genome loading, BED peak processing, sequence encoding, model training, evaluation, and experiment management. Within this framework, we evaluate a hybrid architecture that combines reverse-complement-aware processing, dilated residual convolutional blocks, bidirectional GRU sequence modeling, and gated attention pooling.

Our contributions are:

- A modular genomic sequence framework for controlled regulatory prediction experiments, including preprocessing, training, evaluation, and result tracking.
- An RC-dilated-BiGRU-gated architecture designed to combine strand-awareness, multi-scale convolutional context, recurrent dependency modeling, and attention-based aggregation.
- A two-task empirical evaluation on ATAC and CTCF prediction against classical baselines and state-of-the-art-inspired architectures, including DeepSEA-style, DanQ-style, and Basenji-style models.

The implementation of the proposed system is publicly available.[1]

[1]Code repository: https://github.com/JohnNDaras/DNA_Regulatory_Sequence_Framework. Demonstration video: https://youtu.be/lmixbBNghwM.

2 Related Work

Deep learning has become a powerful approach for predicting regulatory genomic activity directly from DNA sequence. Early successful models demonstrated that one-hot encoded nucleotide sequences can be treated as structured input signals, allowing neural networks to learn sequence motifs and regulatory patterns without manually engineered features. This is especially important in regulatory genomics, where functional sequence elements may depend on both local motifs and their arrangement within a broader genomic context.

DeepSEA [7] is one of the most influential examples of convolutional modeling for regulatory genomics. It uses convolutional neural networks to predict chromatin features from DNA sequence, showing that CNNs can automatically learn motif-like detectors relevant to regulatory activity. This motivates the use of convolutional baselines in our work, since CNNs provide a natural mechanism for identifying local nucleotide patterns.

DanQ [5] extended this direction by combining convolutional layers with recurrent sequence modeling. In this architecture, convolutional filters first detect local motifs, while recurrent layers model dependencies between motifs along the sequence. This is particularly relevant for regulatory DNA, where the order, spacing, and co-occurrence of motifs may influence functional activity. Our DanQ-inspired comparison follows this principle by combining local convolutional feature extraction with bidirectional recurrent modeling.

Basenji [2] and Basenji2 further advanced sequence-to-function prediction by using convolutional architectures with dilated layers, allowing models to capture longer-range genomic context. Dilated convolutions are useful because they expand the receptive field without requiring excessively deep networks or very large computational cost. This idea motivates our Basenji-style baseline and also informs the dilated residual component of our proposed model.

In this work, we do not directly reproduce the official DeepSEA, DanQ, or Basenji/Basenji2 implementations. Those models were originally designed with different input lengths, output spaces, datasets, and training regimes, making direct comparison inappropriate in our binary ATAC and CTCF setting. Instead, we implement state-of-the-art-inspired architectures under a unified experimental pipeline. This allows us to compare the main architectural ideas—pure convolution, convolution plus recurrence, and dilated convolution—under the same preprocessing, training, and evaluation conditions.

3 Methods

3.1 Task Formulation

We formulate regulatory sequence prediction as a binary classification problem. Given a fixed-length DNA window x, the model predicts whether the corresponding genomic region is associated with a regulatory signal. The target label is $y \in \{0, 1\}$, where $y = 1$ denotes a positive regulatory region, such as an ATAC-seq peak or CTCF binding site, and $y = 0$ denotes a negative background region. The model outputs a probability $p(y = 1 \mid x)$, which is later converted into a binary prediction using a validation-selected threshold.

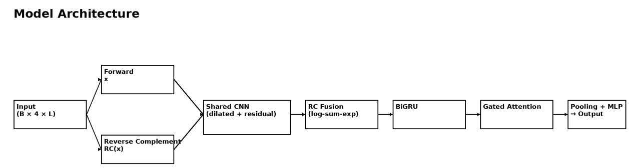

Figure 1: Architecture of the proposed RC-dilated-CNN–BiGRU–gated-attention model. The model processes both the input DNA sequence and its reverse complement using a shared dilated residual convolutional feature extractor. The resulting feature maps are merged via log-sum-exp fusion to enforce reverse-complement consistency. A bidirectional GRU models long-range dependencies, followed by a gated attention mechanism that refines informative regions. The final prediction is obtained through global pooling and a multi-layer perceptron.

Each DNA sequence is represented over the nucleotide alphabet $\{A, C, G, T\}$. Since neural networks require numerical input, sequences are converted into a four-channel one-hot representation, where each position contains a binary vector corresponding to the observed nucleotide. This representation preserves the sequential structure of DNA while allowing convolutional and recurrent models to process the input directly.

3.2 Data and Preprocessing

We evaluate the framework on two regulatory genomics tasks: ATAC-seq chromatin accessibility prediction and CTCF binding prediction. ATAC-seq peaks represent genomic regions with accessible chromatin, while CTCF peaks correspond to binding regions of the CTCF transcription factor. For both tasks, genomic intervals are provided as BED files, and fixed-length sequence windows centered on peak midpoints are extracted from the hg19 reference genome.

To avoid information leakage between training and validation, we use chromosome-based splits rather than random region splits. For the ATAC task, chromosomes 3 and 4 are used for validation, and the remaining chromosomes are used for training. For the CTCF task, chromosomes 2 and 3 are used for validation, chromosome 1 is reserved for testing, and chromosomes 4–22 are used for training. This split strategy evaluates generalization to unseen chromosomes and prevents models from exploiting local genomic similarity.

A key component of the framework is the construction of negative examples. Rather than sampling arbitrary genomic windows, we use an improved peak-centered dataset generator that pairs each positive sequence with multiple negative samples. Negative windows are sampled away from known peaks, constrained to be at least a fixed genomic distance from any annotated peak, and approximately matched to the GC content of the corresponding positive sequence. In addition, windows containing a high fraction of ambiguous bases are filtered out.

This negative sampling strategy reduces trivial discrimination based on sequence composition or low-quality genomic regions and encourages the model to learn biologically meaningful regulatory patterns. During evaluation, a deterministic dataset construction is used to ensure consistency across models, while all models are trained using the same sampling procedure to maintain a fair comparison.

3.3 Proposed Model

The proposed model is a reverse-complement-aware hybrid neural architecture for nucleotide sequence classification. The input sequence is first processed by a convolutional stem that learns local nucleotide patterns. This is followed by a stack of dilated residual convolutional blocks. The residual connections stabilize optimization, while dilated convolutions expand the receptive field and allow the model to capture broader regulatory context without excessive depth.

Because DNA is double-stranded, the same regulatory pattern may appear on either the forward strand or the reverse-complement strand. To account for this, the model processes both the original sequence and its reverse complement using shared feature extraction layers. The two feature representations are then fused using a smooth log-sum-exp operation, producing a strand-aware representation before downstream sequence modeling.

The fused representation is passed through a bidirectional GRU, which models dependencies across sequence positions in both directions. A gated attention refinement module then reweights positional interactions, allowing the network to focus on informative regions within the sequence. Finally, the model combines attention pooling, average pooling, and max pooling summaries and passes the concatenated representation through a multilayer prediction head to produce a single binary logit.

3.4 Training and Evaluation

All models are trained using binary cross-entropy with logits. To account for class imbalance, the positive-class weight is estimated from training batches and used in the loss function. We optimize models with AdamW [3], and use a OneCycle learning-rate schedule to improve convergence. Training also includes label smoothing, mixed-precision computation on GPU, and exponential moving average weights for more stable validation performance.

Early stopping is applied based on validation accuracy. After training, models are evaluated using accuracy, F1-score, AUROC, and AUPRC. Since the optimal classification threshold may differ between models and tasks, the threshold is selected on the validation set to maximize F1-score. All baselines, state-of-the-art-inspired architectures, and the proposed model are trained under the same preprocessing, optimization, and evaluation protocol to ensure a controlled comparison.

4 Experiments

4.1 Baselines

We compare the proposed model against both simple baselines and stronger state-of-the-art-inspired architectures. The simple baselines include logistic regression, a multilayer perceptron, and a standard one-dimensional CNN. These models establish a progression from linear sequence classification to nonlinear feature learning and local motif detection.

In addition, we implement three architectures inspired by established regulatory genomics models. The DeepSEA-style model is a convolutional network designed to capture local motif-like sequence features. The DanQ-style model combines convolutional feature extraction with bidirectional recurrent sequence modeling, allowing it to represent dependencies between detected motifs. The Basenji-style model uses dilated convolutional layers to increase receptive field size and capture broader sequence context. These models are not official reproductions of the original systems; rather, they are controlled implementations of their main architectural principles within our binary prediction framework.

All models are trained using the same data splits, sequence encoding, optimization procedure, and evaluation metrics. This ensures that performance differences primarily reflect architectural choices rather than differences in preprocessing or training protocol.

4.2 Datasets

We evaluate on two regulatory genomics tasks. The first task is ATAC-seq chromatin accessibility prediction, where positive examples correspond to accessible chromatin regions. This task is biologically broad and can involve diverse regulatory signals, making it a challenging benchmark for sequence-based prediction.

The second task is CTCF binding prediction, where positive examples correspond to binding sites of the CTCF transcription factor. Compared with ATAC accessibility, CTCF binding is expected to be more motif-driven, since CTCF has a well-characterized binding pattern. Evaluating on both tasks allows us to test whether the proposed architecture generalizes across different types of regulatory signals.

For both datasets, we use fixed-length DNA windows extracted from the hg19 reference genome and apply chromosome-based splitting to reduce leakage between training and validation regions.

4.3 Evaluation Metrics

We report four metrics: accuracy, F1-score, AUROC, and AUPRC. Accuracy measures the fraction of correct predictions after thresholding. F1-score balances precision and recall and is useful when positive and negative examples may not be equally represented. AUROC measures ranking quality across classification thresholds, while AUPRC emphasizes performance on the positive class and is especially relevant for regulatory genomics tasks where positive regions can be sparse.

For each model, the classification threshold is selected on the validation set to maximize F1-score. The same metric computation is used for all models and both tasks.

5 Results

Table 1 reports the main validation results on the ATAC and CTCF tasks. We include the standard CNN baseline, the three state-of-the-art-inspired architectures, and the proposed model. Simpler baselines such as logistic regression and multilayer perceptrons were also evaluated but are omitted for compactness because they performed substantially worse than the deep sequence models.

The proposed model achieves the best overall performance on both regulatory prediction tasks. This trend is also shown visually in Figure 2. On ATAC, the proposed model obtains the highest

Figure 2: Performance summary across ATAC and CTCF regulatory prediction tasks. (a) ATAC AUPRC comparison across CNN baseline, DeepSEA-style, DanQ-style, Basenji-style, and the proposed model. (b) CTCF AUPRC comparison across the same architectures. (c) AUPRC comparison across both ATAC and CTCF, showing cross-task generalization. (d) AUROC comparison across both tasks. (e) Absolute performance gain of the proposed model over the strongest non-proposed comparator for each metric and task. (f) ATAC metric profile across accuracy, F1-score, AUROC, and AUPRC. (g) CTCF metric profile across the same metrics. (h) Heatmap summary of all main metrics across tasks and architectures. (i) Multi-seed robustness of the proposed model on ATAC, showing stable validation accuracy across random seeds.

AUPRC among all evaluated architectures (Fig. 2a), with a clear improvement over the strongest state-of-the-art-inspired baselines. On CTCF, overall model performance is higher, but the proposed model still achieves the best AUPRC (Fig. 2b).

The grouped comparisons across both tasks further show that the proposed model generalizes beyond a single regulatory setting. In terms of both AUPRC and AUROC, the model remains consistently competitive across ATAC and CTCF (Fig. 2c–d). The absolute improvement over the strongest non-proposed comparator is especially large on ATAC, while the CTCF gains are smaller but still positive across the main metrics (Fig. 2e). This pattern suggests that ATAC is the more challenging task, whereas CTCF is more motif-driven and easier for most deep sequence models.

The metric profiles provide additional support for this interpretation. On ATAC, performance separates more strongly across architectures, with recurrent and dilated models improving substantially over the CNN baseline and the proposed model achieving the strongest overall profile (Fig. 2f). On CTCF, deep models cluster closer together at high performance, but the proposed architecture still leads across the main metrics (Fig. 2g). The heatmap summary confirms the same trend across tasks, models, and metrics (Fig. 2h).

Finally, the multi-seed robustness analysis shows that the proposed model is stable across random initializations on ATAC (Fig. 2i). Validation accuracy remains within a narrow range across seeds 42, 43, and 44, suggesting that the observed performance is not due to a single favorable run.

Table 1: Validation performance on ATAC and CTCF regulatory prediction tasks.

Task	Model	Acc	F1	AUROC	AUPRC
ATAC	CNN baseline	0.801	0.816	0.892	0.885
ATAC	DeepSEA-style	0.789	0.804	0.878	0.871
ATAC	DanQ-style	0.867	0.870	0.939	0.936
ATAC	Basenji-style	0.862	0.865	0.936	0.935
ATAC	Ours	**0.918**	**0.919**	**0.969**	**0.967**
CTCF	CNN baseline	0.946	0.947	0.988	0.987
CTCF	DeepSEA-style	0.950	0.951	0.988	0.988
CTCF	DanQ-style	0.964	0.964	0.988	0.986
CTCF	Basenji-style	0.958	0.958	0.992	0.992
CTCF	Ours	**0.979**	**0.979**	**0.995**	**0.995**

6 Ablation and Robustness

To assess robustness, we repeated the full model on the ATAC task using three random seeds. The validation accuracy remained highly stable across seeds, ranging approximately from 0.918 to 0.919, as shown in Fig. 2i. This indicates that the performance of the proposed model is not dependent on a single favorable initialization and that the training procedure is relatively stable.

We also use the architecture comparisons as a compact ablation of modeling principles. The CNN baseline captures local sequence motifs but lacks explicit long-range or recurrent modeling. The DanQ-style model adds recurrent sequence processing, while the Basenji-style model emphasizes dilated convolutional context. Both improve substantially over the CNN baseline on ATAC, but neither reaches the performance of the full proposed architecture. This suggests that no single component is sufficient by itself; rather, the strongest performance comes from combining reverse-complement-aware processing, dilated residual convolutions, recurrent sequence modeling, and gated attention pooling within one model.

7 Discussion and Limitations

The proposed architecture demonstrates clear and consistent performance gains across both regulatory prediction tasks. Under a controlled experimental setting, it outperforms all baselines and state-of-the-art-inspired models by a meaningful margin, particularly on the more challenging ATAC task. This suggests that the combination of reverse-complement-aware processing, dilated residual convolutions, recurrent sequence modeling, and gated attention captures complementary aspects of regulatory sequence structure that are not fully exploited by existing architectures when used in isolation.

An important strength of this work is the strictly controlled comparison framework. All models are trained using identical preprocessing, chromosome-based splits, optimization strategies, and evaluation protocols. This isolates architectural contributions and provides a reliable assessment of how different modeling choices impact performance. The consistent improvements observed across both ATAC and CTCF tasks indicate that the proposed design is not task-specific, but instead provides a general and robust modeling approach for regulatory genomics.

The improved negative sampling strategy also plays a critical role in the strength of the benchmark. By controlling peak distance, GC content, and ambiguous bases, the framework reduces shortcut learning and forces models to rely on more informative sequence features. This contributes to a more realistic and challenging evaluation setting compared to naive random negative sampling.

At the same time, we emphasize that the comparison is performed against state-of-the-art-inspired architectures rather than official pretrained models. This design choice ensures fairness and reproducibility within our experimental setup, but it does not directly position the model against large-scale pretrained systems trained on broader multi-task genomic datasets. Extending the framework to such settings is a natural next step.

Future work can build on these results in several directions. Scaling to larger and more diverse genomic datasets would test the model's ability to generalize across cell types and regulatory contexts. Integrating official benchmark datasets would further strengthen comparability with existing literature. In addition, improving interpretability remains an important goal, as understanding which sequence features drive predictions is critical for biological insight and downstream applications.

8 Conclusion

We introduced a modular framework for genomic sequence modeling and a hybrid RC-dilated-BiGRU-gated architecture for regulatory prediction. Across both ATAC chromatin accessibility and CTCF binding tasks, the proposed model consistently outperforms strong baselines and state-of-the-art-inspired architectures under a unified and controlled evaluation setting. The results highlight the effectiveness of combining strand-awareness, multi-scale convolutional context, recurrent modeling, and attention mechanisms for DNA sequence analysis, and suggest that this integrated approach is a promising direction for future work in regulatory genomics.

References

[1] Jason D. Buenrostro, Paul G. Giresi, Lisa C. Zaba, Howard Y. Chang, and William J. Greenleaf. 2013. Transposition of native chromatin for fast and sensitive epigenomic profiling of open chromatin, DNA-binding proteins and nucleosome position. *Nature Methods* 10, 12 (2013), 1213–1218. doi:10.1038/nmeth.2688

[2] David R. Kelley, Yakir A. Reshef, Max Bileschi, David Belanger, Cory Y. McLean, and Jasper Snoek. 2018. Sequential regulatory activity prediction across chromosomes with convolutional neural networks. *Genome Research* 28, 5 (2018), 739–750. doi:10.1101/gr.227819.117

[3] Ilya Loshchilov and Frank Hutter. 2019. Decoupled Weight Decay Regularization. In *International Conference on Learning Representations*. https://openreview.net/forum?id=Bkg6RiCqY7 Published as a conference paper at ICLR 2019.

[4] Chin-Tong Ong and Victor G. Corces. 2014. CTCF: an architectural protein bridging genome topology and function. *Nature Reviews Genetics* 15, 4 (2014), 234–246. doi:10.1038/nrg3663

[5] Daniel Quang and Xiaohui Xie. 2016. DanQ: a hybrid convolutional and recurrent deep neural network for quantifying the function of DNA sequences. *Nucleic Acids Research* 44, 11 (2016), e107. doi:10.1093/nar/gkw226

[6] Hannah Zhou, Avanti Shrikumar, and Anshul Kundaje. 2022. Towards a Better Understanding of Reverse-Complement Equivariance for Deep Learning Models in Genomics. In *Proceedings of the 16th Machine Learning in Computational Biology Meeting (Proceedings of Machine Learning Research, Vol. 165)*. PMLR, 1–33. https://proceedings.mlr.press/v165/zhou22a.html

[7] Jian Zhou and Olga G. Troyanskaya. 2015. Predicting effects of noncoding variants with deep learning-based sequence model. *Nature Methods* 12, 10 (2015), 931–934. doi:10.1038/nmeth.3547